Quiz II

Test your knowledge of different systems that you have learned in this course by completing this quiz.

Quiz ?

Statement

In Google File System (GFS), when a client needs to access data, it contacts the master and gets the data from the master.

Select Multiple Options

☐ True

☐ False

Check Answer

Quiz ?

Statement

The main responsibility of the GFS master is:

Select Multiple Options

☐ To coordinate all read and write requests from clients.

☐ To balance the storage load between different ChunkServers.

☐ To coordinate concurrent write requests to a file from multiple clients.

☐ Store all the filesystem metadata, like file names, total parts, locations of the parts, etc.

Check Answer

Quiz ?

Statement

When scaling systems, mostly distributed system give up:

Select Multiple Options

☐ Availability in favor of Partition-tolerance

☐ Availability in favor of Consistency

☐ Atomicity in favor of Availability

☐ Data consistency in favor of availability

Check Answer

Quiz ?

Statement

The motivation for an eventual consistency model was:

Select Multiple Options

☐ It is impossible to have highly available replicated data that is fully consistent in a system that can survive network partitioning.

☐ Data inconsistencies are highly undesirable, so the primary focus should be on ensuring that all replicas are consistent.

☐ Distributed commit protocols can never work reliably, so data is bound to become inconsistent on some participants.

☐ Data might be in an inconsistent state during the execution of transactions but will be made consistent when they commit.

Check Answer

Quiz ?

Statement

In BigTable, a Tablet refers to:

Select Multiple Options

☐ A subset of rows and column families

☐ A contiguous range of rows of a table

☐ All columns of contiguous rows of a Locality Group

☐ All columns families of contiguous rows of a table

Check Answer

Quiz ?

Statement

Split-brain is a condition that arises when:

Select Multiple Options

☐ A leader election is taking place.

☐ A network is partitioned.

☐ The election algorithm fails to elect a leader.

☐ The leader initiates a leader election.

Check Answer

Quiz ?

Statement

The main difference between the Google File System (GFS) and the Hadoop Distributed File System (HDFS) is that HDFS:

Select Multiple Options

☐ HDFS stores all filesystem metadata on a single master.

☐ HDFS stores files in fixed-size blocks.

☐ HDFS does not allow concurrent reading of a file.

☐ In HDFS, there are no concurrent writers and arbitrary file modifications.

Check Answer

Quiz ?

Statement

Distributed systems use the following techniques for Fencing. Select all that apply.

Select Multiple Options

☐ Resource Fencing

☐ Network Fencing

☐ Node Fencing

☐ None of the above

Check Answer

Quiz ?

Statement

In distributed systems, Fencing is used to:

Select Multiple Options

☐ To isolate a central server from other servers.

☐ To isolate a server while it initializes.

☐ Isolate a faulty server to prevent it from doing any damage or causing corruption.

☐ To enhance security around certain critical nodes.

Check Answer

Quiz ?

Statement

A column family in BigTable is:

Select Multiple Options

☐ A group of columns within a single table.

☐ A group of related columns among multiple tables.

☐ A set of columns that are shared across multiple tables.

☐ A preconfigured set of columns to make it easier to set up a Bigtable instance.

Check Answer

Quiz ?

Statement

Dynamo uses vector clocks to:

Select Multiple Options

☐ To garbage collect older versions of an object.

☐ To store multiple versions of data.

☐ Determine causality between different updates on the same object.

☐ To serialize concurrent updates to an object.

Check Answer

Input your markdown text here

Quiz ?

Statement

In distributed systems, replication gives following benefits. Select all that apply.

Select Multiple Options

☐ Availability

☐ Consistency

☐ Reliability

☐ Better read performance

Check Answer

Quiz ?

Statement

In distributed systems, Heartbeat is used to:

Select Multiple Options

☐ Detect dead servers

☐ Synchronize operations between servers

☐ Measure load on servers

☐ Transfer configuration changes

Check Answer

Quiz ?

Statement

What statements are true about the Google File System (GFS)? Select all that apply.

Select Multiple Options

☐ Different parts of a file can be stored on different servers.

☐ GFS can store multiple replicas of a file's data on different servers.

☐ Any file stored in GFS can be bigger than the capacity of a server.

☐ Transfer configuration changes

Check Answer

Quiz ?

Statement

**What is the importance of choosing a good row key in BigTable? **

Select Multiple Options

☐ Take advantage of the locality of adjacent rows.

☐ Avoid the use of duplicate keys.

☐ Enable rapid search for an arbitrary row based on its key.

☐ Make queries user-friendly for easier debugging.

Check Answer

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